

$$x(n) - 2x(n-1) = n \cdot 2^n \cdot u(n) + \delta(n)$$

$$x(n) = 0, \quad n < 0$$

$$y(n) = \frac{x(n)}{2^n}$$

$$2 \frac{x(n-1)}{2^n} = \frac{x(n-1)}{2^{n-1}} = y(n-1)$$

$$\rightarrow y(n) - y(n-1) = n \cdot u(n) + \delta(n)$$

Summing up LHS & RHS:

$$y(n) - y(n-1) = \frac{n(n+1)}{2} u(n) + \delta(n)$$

$$y(n) = \left[1 + \frac{n(n+1)}{2} \right] \cdot u(n)$$

$$x(n) = 2^n \cdot y(n)$$

$$= 2^{n-1} (n^2 + n + 2) \cdot u(n)$$

$\rightarrow n^2 \cdot 2^{n-1}$ term dominates - so $x(n) = \mathcal{O}(n^2 \cdot 2^n)$